How the AI traffic analyser works

Project implementation snapshot | 24 September 2026

The app combines image and video processing (IVP), pretrained neural networks, tracking, and explicit decision rules. It runs locally and accepts uploaded images or videos. It is a working experimental prototype, with several violation categories still awaiting reliable scene interpretation and real traffic validation.

The processing flow

Step	What happens
1. Prepare footage	Correct orientation, optionally resize and enhance the colour frame; preserve timestamps and coordinate mappings.
2. Detect objects	Locate people, vehicles, traffic lights and helmet / bare-head candidates using trained detectors.
3. Understand the scene	Estimate road and sidewalk regions; propose markings and signs; extract geometric features.
4. Connect evidence	Associate riders with motorcycles, link observations across video frames, and estimate applicable road approaches.
5. Evaluate rules	Check the evidence required by each rule. Missing or ambiguous context produces cannot_assess.
6. Present results	Show cyan object detections and green reported violation boxes; save evidence, settings and structured reports.

A useful explanation for class

"IVP prepares the pixels and extracts features. AI identifies objects and regions. Tracking connects observations over time. Our rule engine combines this evidence to decide whether a possible violation can be reported."

Key distinction: detecting a motorcycle is not detecting a violation. A violation needs additional evidence, such as an associated bare head, a prohibited movement, or a crossing while the relevant signal is red.

Models and local execution

Different models answer different visual questions.

Component	Model / method	Role and present limitation
Road users	Ultralytics YOLO11s	Detects people, bicycles, cars, motorcycles, buses, trucks, traffic lights and stop signs. Does not directly recognise all nine violations.
Helmet evidence	YOLOv8n helmet detector iam-tsr/yolov8n-helmet-detection	Predicts helmet and bare-head boxes. Small, hidden or unfamiliar heads can be missed.
Scene proposals	Grounding DINO Tiny	Uses prompts for crosswalks, stop lines, sidewalks, lights, signs and road arrows. Proposals require further checks.
Road / sidewalk	SegFormer B0, Cityscapes fine-tune	Labels pixels as road or sidewalk. These masks do not establish legal road usage.
Traffic signs	ViT GTSRB classifier bazyl/gtsrb-model	Classifies cropped sign candidates from a limited vocabulary. Explicit turn-prohibition coverage is incomplete.
Video identity	ByteTrack	Links road-user detections across frames. IDs can be lost or swapped during occlusion.

Where computation happens

Inference runs on this computer. YOLO and Grounding DINO use the available NVIDIA GPU through PyTorch/CUDA. SegFormer and the sign classifier currently run on CPU. Model weights download when needed and are cached; the inference pipeline does not require uploading footage to a hosted AI API.

The AI runtime uses a separate Python 3.11 environment. Streamlit provides the browser interface; OpenCV, NumPy and Pillow handle image operations; PyAV and FFmpeg support video decoding and export.

Pretrained does not mean locally validated

We have not fine-tuned these models on a project-specific Indian traffic dataset. COCO, Cityscapes and GTSRB are general or non-Indian benchmarks. Crowding, auto-rickshaws, helmet designs, camera angles and local signage can differ from their training data. Model confidence is not a calibrated probability of a traffic violation.

From pixels to evidence

IVP and AI cooperate, while keeping colour and geometry traceable.

Image preparation and colour handling

The colour pipeline applies optional denoising, gamma correction, CLAHE and sharpening after orientation correction and resizing. Features such as edges, masks, contours and lines branch separately. AI road-user detection uses the processed colour frame; scene and signal analysis use an unenhanced frame resized to the same dimensions, reducing colour distortion from enhancement.

Arrays are kept as RGB internally and converted to BGR where required by OpenCV or the YOLO interface. Output coordinates refer to the processed image, with a saved mapping back to the display-oriented source. Video records retain timestamps so a decision can be traced to its frame.

Riders and helmets

The app first detects a motorcycle and nearby people. Geometric checks test whether each person plausibly occupies the motorcycle seat region and has an appropriate size and position. Ambiguous overlaps are rejected. Helmet / bare-head boxes must match the upper region of a uniquely associated person.

For video, repeated person-motorcycle observations and similar motion strengthen the association. A missing helmet box alone never proves helmet absence: the rule needs positive bare-head evidence. Counting three unrelated people near a motorcycle is also insufficient.

Road context and controlling signals

Grounding DINO proposes locations. Classical processing checks white stripe patterns, line segments and arrow geometry, while segmentation estimates road and sidewalk masks. Arrow and stop-line geometry can form a candidate road approach. Signs, crossings and lights are associated with an approach only when current geometric checks resolve a match.

A red colour region is only a colour observation. The app must also establish a traffic-light detection, its state over time, and a credible association with the vehicle approach. These automatic associations remain experimental.

Video provides information a photograph cannot

ByteTrack IDs and timestamped positions support line crossings, stopping, movement direction and entry transitions. Camera-motion checks help avoid treating camera movement as vehicle movement. Scene changes reset accumulated context. A single image cannot demonstrate that a vehicle crossed a stop line during red.

What the nine checks require

Rule logic exists for nine categories; end-to-end reliability is not yet established.

Category	Evidence required	Current qualification
Helmet absence	Motorcycle + associated rider + confident bare-head detection	Works on some still-image examples; crowded and occluded cases remain uncertain.
Triple riding	At least three distinct riders associated with one motorcycle	Geometric and temporal association implemented; real positive recall remains unverified.
Red-light crossing	Tracked crossing of a finite stop line while the associated signal remains red	Video only; reliable automatic signal and approach association still needs validation.
Stop-line encroachment	Observed entry during red, then sustained stopping beyond the line	Video history required. Being first seen beyond a line is insufficient.
Crossing obstruction during red	Sustained stationary occupancy of an approach-linked crossing during red	Crossing association is checked; real scene validation remains incomplete.
Wrong-way movement	Tracked movement opposite an independently supported permitted direction	Dominant traffic flow is not treated as the legal direction. Arrow extraction is experimental.
Footpath riding	Moving two-wheeler remains inside a sidewalk region	Needs stable video and trustworthy segmentation; masks can be wrong.
Restricted entry	Recognised applicable restriction and a tracked outside-to-inside transition	Limited sign vocabulary and unverified road applicability restrict coverage.
Prohibited turn	Applicable prohibition plus a correctly interpreted vehicle turn	Unfinished: reliable turn-sign recognition and automatic road-plane calibration are missing.

Three possible outcomes

violation: the implemented checks found supporting evidence.
no_violation_observed: no violation was observed for the evaluated evidence; this is not a universal clearance.
cannot_assess: necessary evidence is missing, uncertain or unsupported.

Green boxes represent reported experimental rule outcomes, not ground-truth enforcement decisions. The current system must not be presented as a validated nine-rule detector.

Outputs, evidence and next work

The prototype is designed to make its decisions inspectable.

What the app saves

Each run has its own settings and outputs. Image results include detection and violation previews. Video results retain playback timing and source audio when available, with browser-compatible H.264 export. Structured records include timestamps, object boxes, track IDs, rule verdicts, reasons, scene evidence and model identifiers. The original upload remains intact.

The event layer confirms persistent video observations and avoids repeatedly reporting the same event. Short transition events, such as crossing a line, are handled separately so they are not lost. Cancellation and incremental video processing keep jobs manageable.

What has actually been checked

The project has unit and integration checks for rule logic, geometry, exports and job handling, plus selected real-image experiments. A rider-association revision removed three known false triple-riding reports in the recorded development cases. This is a useful regression check, not a measured accuracy score or proof of positive-case recall.

A separate Vietnamese turn-sign detector was evaluated because its labels include no-left, no-right and no-U-turn signs. It failed the tested no-left-turn photograph and illustration, so it is not connected to automatic violation decisions. Lowering thresholds would not correct its wrong labels.

Next implementation priorities

1. Evaluate a better explicit turn-sign model or collect representative labelled data.
2. Improve automatic road geometry and controlling-signal association.
3. Validate rider association and helmets on crowded Indian traffic.
4. Test every category using real positive, negative and ambiguous videos, reporting precision, recall and failure cases.

Sources and implementation references

This document describes the local implementation, principally traffic_ai/analyzer.py, perception.py, riders.py, approaches.py, rules.py, worker.py and the validation notes in docs/. Model references below identify the components; their upstream results are not project accuracy measurements.

YOLO and tracking documentation
Helmet checkpoint
Grounding DINO Tiny
SegFormer checkpoint
GTSRB sign classifier